Giulio **De Pasquale, PhD**

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Low-level security and offensive research on agentic frameworks; declarative tooling with Nix, open source contributor.

Work Experience

Senior Security Researcher - CoreOS
Roku — *Cambridge, United Kingdom* — *June 2025 - Present*

Lead **offensive security research on AI agents**, developing attack techniques against LLM-driven agentic systems. Conduct **firmware and bootloader analysis** across Roku's device ecosystem, static and dynamic, at enterprise scale.

https://roku.com

Security Researcher
Apple — *London, United Kingdom* — *October 2023 - June 2025*

Worked on **DAST infrastructure** in Golang with an extensible plugin system for custom detections, scanning Apple's internal systems and leading its CI/CD integration.

https://apple.com

Visiting PhD Student
University of California, Santa Barbara — *Santa Barbara, California, USA* — *June 2022 - October 2023*

Applied **AI planning** to post-exploitation, deriving system facts and planning privilege escalation action chains, as part of the DARPA HARDEN program.

https://seclab.cs.ucsb.edu | Advised by *Prof. Christopher Kruegel and Prof. Giovanni Vigna*

Research Internship
Apple — *London, United Kingdom* — *May 2022 - June 2022*

Internship in the Applied Research team of Apple Information Security (AIS).
Developed an automated firmware analysis tool that uncovered a **zero-day vulnerability**.

https://apple.com | Advised by *Andre Protas*

Visiting PhD Student
University College of London — *London, United Kingdom* — *June 2021 - February 2024*

Joined UCL's ISec research group, collaborating on research projects and coordinating fortnightly **hands-on seminars** on information security and CTFs.

https://sec.cs.ucl.ac.uk

Research Internship
Tweag I/O — *Remote* — *June 2021 - September 2021*

Improved the robustness of the type system of **Nickel**, a declarative configuration language written in Rust.

https://github.com/tweag/nickel | Advised by *Dr. Yan Hamdaoui*

Research Internship - NExT Special Projects
Microsoft Research — *Redmond, Washington, USA* — *June 2019 - September 2019*

Researched malware analysis techniques to detect **cryptomining** and other active threats in ELF core files for *Microsoft Project Freta*, developed in Rust.

https://aka.ms/freta | Advised by *Mike Walker*

Visiting Graduate Researcher
University of California, Santa Barbara — *Santa Barbara, California, USA* — *January 2018 - July 2018*

Researched **deobfuscation** techniques for software protected by VM obfuscation and fuzzing methodologies targeting IoT devices.

https://seclab.cs.ucsb.edu | Advised by *Prof. Christopher Kruegel and Prof. Giovanni Vigna*

Publications

ChainReactor: Automated Privilege Escalation Chain Discovery via AI Planning — *Philadelphia, Pennsylvania, USA*

33rd USENIX Security Symposium *(USENIX '24)* — *August 2024*

Models exploitation as **AI planning** to discover privilege escalation chains on Unix, awarded the Distinguished Artifact Award.

- **Authors:**
  Giulio De Pasquale, *Ilya Grishchenko, Riccardo Iesari, Gabriel Pizarro, Lorenzo Cavallaro, Christopher Kruegel and Giovanni Vigna.*

ROPfuscator: Robust Obfuscation with ROP — *San Francisco, California, USA*

17th IEEE Workshop on Offensive Technologies *(WOOT '23)* — *May 2023*

A compiler-driven **obfuscation pass** over ROP binaries protecting software from man-at-the-end attacks and reverse engineering.

- **Authors:**
  Giulio De Pasquale, *Fukutomo Nakanishi, Daniele Ferla, Lorenzo Cavallaro.*

DIANE: Identifying Fuzzing Triggers in Apps to Generate Under-constrained Inputs for IoT Devices — *Online*

42nd IEEE Symposium on Security and Privacy *(S&P)* — *May 2021*

Combines static and dynamic analysis to find **fuzzing triggers** and automatically fuzz IoT devices.

- **Authors:**
  Nilo Redini, Andrea Continella, Dipanjan Das, ***Giulio De Pasquale****, Noah Spahn, Aravind Machiry, Antonio Bianchi, Christopher Kruegel, Giovanni Vigna.*

ShieldFS: A Self-healing, Ransomware-aware Filesystem — *Los Angeles, California, USA*

32nd Annual Computer Security Applications Conference *(ACSAC)* & Black Hat '17 — *December 2016*

Detects **ransomware** at runtime from filesystem activity, adapting detection models to usage habits.

- **Authors:**
  Andrea Continella, Alessandro Guagnelli, Giovanni Zingaro, ***Giulio De Pasquale****, Alessandro Barenghi, Stefano Zanero, Federico Maggi.*

Education

Ph.D Degree in Computer Security — *London, United Kingdom*

King's College London, Department of Informatics — *November 2018 - February 2024*

Specialized in automated privilege escalation, program analysis, software obfuscation, and IoT firmware analysis. Researched automated exploitation chain discovery and resilience against reverse engineering and ransomware attacks.

https://s2lab.kcl.ac.uk | Advised by *Prof. Lorenzo Cavallaro*

B.Sc Degree in Engineering of Computing Systems — *Milan, Italy*

Politecnico di Milano — *September 2011 - September 2017*

Covered Algorithms, Operating Systems, Databases, and Computer Architectures, alongside Engineering subjects including Physics, Math, and Algebra.

Invited Talks

Coherent, Therefore Trusted — *September 2026*
Roku Engineering Symposium · Aarhus, Denmark

Reverse Engineering to AI: A Journey Through Modern Security Research — *April 2026*
ISC2 East of England Chapter · Remote

(Still) Researching My Way Forward: Playing (CTFs) through PhD to Profession — *February 2026*
UCL Electronic and Electrical Engineering · London, United Kingdom

Researching My Way Forward: Playing (CTFs) through PhD to Profession — *February 2025*
UCL Electronic and Electrical Engineering · London, United Kingdom

Selected Personal Projects

Pasticciotto https://github.com/peperunas/pasticciotto

A **polymorphic Virtual Machine** that protects proprietary code by embedding it into applications, released as a C++ reverse engineering challenge at PoliCTF '17.

Injectopi https://github.com/peperunas/injectopi

A collection of tutorials demonstrating **code injection** techniques on Microsoft Windows, written in C.

AESTracer

A Microsoft Windows driver that scans running processes for **AES keyschedules**, included in the ShieldFS research project published at ACSAC '16.

Conferences Contributions

Program Committee	*USENIX* 2019, *WOOT* 2023, *ESORICS* 2023
Artifact Evaluation Committee	*USENIX* 2021/2022/2023/2024, *NDSS* 2023, *WOOT* 2023

Tech Stack

Programming	Python, Rust, Golang, Nix, C, Souffle, PDDL, TypeScript, Bash, C++, Java
Interests	Reverse engineering, program analysis, system administration, declarative, low-level and systems' programming

CTF Teams



mHACKeroni *Italy*

Italian Capture the Flag (CTF) Team *2019 - 2023*

mHACKeroni is composed by five CTF Italian teams. We qualified and placed 5th to **DEFCON 2019 Finals**. Involved in the organization of the **mHACKeCTF '20** competition.

https://mhackeroni.it

Phish 'n' Chips *London, United Kingdom*

King's College London Capture the Flag (CTF) Team *2018 - 2023*

Founder of the team.

https://phishnchips.co.uk

Shellphish *Santa Barbara, California, USA*

UCSB's Capture the Flag (CTF) Team *2018 - 2023*

Involved in the organization of the **UCSB's iCTF 2018** competition.

https://ictf.cs.ucsb.edu/



Tower of Hanoi *Milan, Italy*

Politecnico di Milano's Capture the Flag (CTF) Team *2014 - 2023*

Involved in the organization of the **PoliCTF 2015** competition held during DIMVA 2015 in Milan.

https://polictf.it